Smart Plant Emotions

1. Introduction

The Smart Plant Emotions System is an IoT-based plant monitoring project using NodeMCU (ESP8266). It monitors soil moisture, light intensity, and temperature, and displays the plant's "emotional state" on a 128×64 OLED screen.

Instead of showing raw values, it represents plant health using visual expressions like Happy, Sad, Sleep, and Hot.

2. Aim

To develop a smart system that:

- Monitors soil moisture
- Detects day/night using LDR
- Measures temperature using DHT11
- Displays plant emotions on OLED

3. Hardware Components

- NodeMCU (ESP8266)
- SH1106 128×64 OLED (I2C)
- Soil Moisture Sensor
- LDR Sensor
- DHT11 Temperature Sensor
- Jumper wires
- 3.3V power supply

4. Working Principle

The system continuously reads sensor data and decides the plant's emotion based on conditions:

- Temperature > 30°C → Hot Face
- Low Light → Sleep Face
- Wet Soil (Daytime) → Happy Face
- Dry Soil (Daytime) → Sad Face

The OLED updates every second.

5. Circuit Connections

OLED → NodeMCU

- VCC → 3.3V
- GND → GND
- SCL → D1
- SDA → D2

Soil Moisture → D3

LDR → D4

DHT11 → D5

⚠ Use only 3.3V with NodeMCU.

6. Features

- Real-time plant status
- Visual emotion display
- Automatic day/night detection
- Temperature alert
- Simple and low-cost design

7. Applications

- Home gardening
- Smart agriculture demos
- School IoT projects
- Indoor plant monitoring

Code:

```
// ---> Libraries
#include <Wire.h>
```

```
#include <U8g2lib.h>
#include <Arduino.h>
#include <DHT.h>

// ---> Pin connections
#define SCL_OLED D1  // SCL pin for NodeMCU
#define SDA_OLED D2  // SDA pin for NodeMCU
#define SOIL_MOISTURE_SENSOR_PIN D3  // Digital pin for soil moisture sensor
#define LDR_SENSOR_PIN D4  // Digital pin for LDR sensor
#define DHT_SENSOR_PIN D5  // Digital pin for DHT11 sensor
#define DHTTYPE DHT11  // DHT11 sensor type

//[page buffer, size = 128 bytes]
U8G2_SH1106_128X64_NONAME_F_HW_I2C OLED(U8G2_R0, /* reset=*/ U8X8_PIN_NONE);

// 'happy', 50x50px
const unsigned char epd_bitmap_happy [] PROGMEM = {
	0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
	0xf0, 0x3f, 0x00, 0x00, 0x00, 0x00, 0x00, 0xfe, 0xff, 0x01, 0x00, 0x00, 0x00, 0x80, 0xff, 0xff,
	0x07, 0x00, 0x00, 0x00, 0xe0, 0xff, 0xff, 0x1f, 0x00, 0x00, 0x00, 0xf8, 0xff, 0xff, 0x7f, 0x00,
	0x00, 0x00, 0xfc, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0xfe, 0xff, 0xff, 0xff, 0x01, 0x00, 0x00,
	0xff, 0xff, 0xff, 0xff, 0x03, 0x00, 0x80, 0xff, 0xff, 0xff, 0xff, 0x07, 0x00, 0xc0, 0xff, 0xff,
	0xff, 0xff, 0x0f, 0x00, 0xc0, 0xff, 0xff, 0xff, 0xff, 0x0f, 0x00, 0xe0, 0xff, 0xff, 0xff, 0xff,
	0x1f, 0x00, 0xe0, 0xff, 0xff, 0xff, 0xff, 0x1f, 0x00, 0xf0, 0xff, 0xff, 0xff, 0xff, 0x3f, 0x00,
	0xf0, 0xff, 0xff, 0xff, 0xff, 0x3f, 0x00, 0xf8, 0xff, 0xff, 0xff, 0xff, 0x7f, 0x00, 0xf8, 0x7f,
```

0xf8, 0x7f, 0xf8, 0x7f, 0x00, 0xf8, 0x3f, 0xf0, 0x3f, 0xf0, 0x7f, 0x00, 0xfc, 0x3f, 0xf0, 0x3f,
0xf0, 0xff, 0x00, 0xfc, 0x3f, 0xf0, 0x3f, 0xf0, 0xff, 0x00, 0xfc, 0x3f, 0xf0, 0x3f, 0xf0, 0xff,
0x00, 0xfc, 0x7f, 0xf8, 0x7f, 0xf8, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc,
0xff, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc, 0xff, 0xff,
0xff, 0xff, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc, 0xe7, 0xff, 0xff, 0x9f,
0xff, 0x00, 0xf8, 0xc7, 0xff, 0xff, 0x8f, 0x7f, 0x00, 0xf8, 0x8f, 0xff, 0xff, 0xc7, 0x7f, 0x00,
0xf8, 0x1f, 0xff, 0xff, 0xe3, 0x7f, 0x00, 0xf0, 0x3f, 0xfe, 0xff, 0xf1, 0x3f, 0x00, 0xf0, 0x7f,
0xf8, 0x7f, 0xf8, 0x3f, 0x00, 0xe0, 0xff, 0xc1, 0x0f, 0xfe, 0x1f, 0x00, 0xe0, 0xff, 0x07, 0x80,
0xff, 0x1f, 0x00, 0xc0, 0xff, 0x1f, 0xe0, 0xff, 0x0f, 0x00, 0xc0, 0xff, 0xff, 0xff, 0xff, 0x0f,
0x00, 0x80, 0xff, 0xff, 0xff, 0xff, 0x07, 0x00, 0x00, 0xff, 0xff, 0xff, 0xff, 0x03, 0x00, 0x00,
0xfe, 0xff, 0xff, 0xff, 0x01, 0x00, 0x00, 0xfc, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0xf8, 0xff,
0xff, 0x7f, 0x00, 0x00, 0x00, 0xe0, 0xff, 0xff, 0x1f, 0x00, 0x00, 0x00, 0x80, 0xff, 0xff, 0x07,
0x00, 0x00, 0x00, 0x00, 0xfe, 0xff, 0x01, 0x00, 0x00, 0x00, 0x00, 0xf0, 0x3f, 0x00, 0x00, 0x00,
0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00
};

// 'sad', 50x50px
const unsigned char epd_bitmap_sad [] PROGMEM = {

0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,

0xf0, 0x3f, 0x00, 0x00, 0x00, 0x00, 0x00, 0xfe, 0xff, 0x01, 0x00, 0x00, 0x00, 0x80, 0xff, 0xff,

0x07, 0x00, 0x00, 0x00, 0xe0, 0xff, 0xff, 0x1f, 0x00, 0x00, 0x00, 0xf8, 0xff, 0xff, 0x7f, 0x00,

0x00, 0x00, 0xfc, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0xfe, 0xff, 0xff, 0xff, 0x01, 0x00, 0x00,

0xff, 0xff, 0xff, 0xff, 0x03, 0x00, 0x80, 0xff, 0xff, 0xff, 0xff, 0x07, 0x00, 0xc0, 0xff, 0xff,

0xff, 0xff, 0x0f, 0x00, 0xc0, 0xff, 0xff, 0xff, 0xff, 0x0f, 0x00, 0xe0, 0x1f, 0xf8, 0x7f, 0xe0,

0x1f, 0x00, 0xe0, 0x07, 0xe0, 0x1f, 0x80, 0x1f, 0x00, 0xf0, 0xc3, 0xc3, 0x0f, 0x0f, 0x3f, 0x00,

0xf0, 0xf3, 0xcf, 0xcf, 0x3f, 0x3f, 0x00, 0xf8, 0xff, 0xff, 0xff, 0xff, 0x7f, 0x00, 0xf8, 0xff,

0xff, 0xff, 0xff, 0x7f, 0x00, 0xf8, 0xff, 0xff, 0xff, 0xff, 0x7f, 0x00, 0xfc, 0xef, 0xff, 0xff,

0xff, 0xff, 0x00, 0xfc, 0xe3, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc, 0xf1, 0xff, 0xff, 0xff, 0xff,

0x00, 0xfc, 0xf0, 0x3f, 0xf0, 0xff, 0xff, 0x00, 0xfc, 0xf0, 0x07, 0x80, 0xff, 0xff, 0x00, 0x7c,

0xf0, 0x03, 0x00, 0xff, 0xff, 0x00, 0x7c, 0xf0, 0x01, 0x00, 0xfe, 0xff, 0x00, 0x7c, 0xf0, 0x00,

0x00, 0xfc, 0xff, 0x00, 0x7c, 0xf0, 0x00, 0x00, 0xfc, 0xff, 0x00, 0xfc, 0x78, 0x00, 0x00, 0xf8,

0xf7, 0x00, 0xf8, 0x7f, 0x00, 0x00, 0xf8, 0x67, 0x00, 0xf8, 0x7f, 0x00, 0x00, 0xf8, 0xc7, 0x00,

0xf8, 0x3f, 0x00, 0x00, 0xf0, 0xc7, 0x00, 0xf0, 0x3f, 0xfc, 0xff, 0xf0, 0x87, 0x01, 0xf0, 0xff,

0xff, 0xff, 0xff, 0x87, 0x01, 0xe0, 0xff, 0xff, 0xff, 0xff, 0x87, 0x01, 0xe0, 0xff, 0xff, 0xff,

0xff, 0x87, 0x01, 0xc0, 0xff, 0xff, 0xff, 0xff, 0xcf, 0x01, 0xc0, 0xff, 0xff, 0xff, 0xff, 0xff,

```
	0x00, 0x80, 0xff, 0xff, 0xff, 0xff, 0x7f, 0x00, 0x00, 0xff, 0xff, 0xff, 0xff, 0x03, 0x00,
0x00,
	0xfe, 0xff, 0xff, 0xff, 0x01, 0x00, 0x00, 0xfc, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0xf8,
0xff,
	0xff, 0x7f, 0x00, 0x00, 0x00, 0xe0, 0xff, 0xff, 0x1f, 0x00, 0x00, 0x00, 0x80, 0xff, 0xff,
0x07,
	0x00, 0x00, 0x00, 0x00, 0xfe, 0xff, 0x01, 0x00, 0x00, 0x00, 0x00, 0xf0, 0x3f, 0x00,
0x00, 0x00,
	0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00
};

// 'sleep', 50x50px
const unsigned char epd_bitmap_sleep [] PROGMEM = {
	0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00,
	0xf0, 0x3f, 0x00, 0x00, 0x00, 0x00, 0x00, 0xfe, 0xff, 0x01, 0x00, 0x00, 0x00, 0x80, 0xff,
0xff,
	0x07, 0x00, 0x00, 0x00, 0xe0, 0xff, 0xff, 0x1f, 0x00, 0x00, 0x00, 0xf8, 0xff, 0xff, 0x7f,
0x00,
	0x00, 0x00, 0xfc, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0xfe, 0xff, 0xff, 0xff, 0x01, 0x00,
0x00,
	0xff, 0xff, 0xff, 0xff, 0x03, 0x00, 0x80, 0xff, 0xff, 0xff, 0xff, 0x07, 0x00, 0xc0, 0xff, 0xff,
	0xff, 0xff, 0x0f, 0x00, 0xc0, 0xff, 0xff, 0xff, 0xff, 0x0f, 0x00, 0xe0, 0xff, 0xff, 0xff, 0xff,
	0x1f, 0x00, 0xe0, 0xff, 0xff, 0xff, 0xff, 0x1f, 0x00, 0xf0, 0xff, 0xff, 0xff, 0xff, 0x3f, 0x00,
	0xf0, 0xff, 0xff, 0xff, 0xff, 0x3f, 0x00, 0xf8, 0xff, 0xff, 0xff, 0xff, 0x7f, 0x00, 0xf8, 0xff,
	0xff, 0xff, 0xff, 0x7f, 0x00, 0xf8, 0xff, 0xff, 0xff, 0xff, 0x7f, 0x00, 0xfc, 0xff, 0xff, 0xff,
	0xff, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc, 0xf3, 0xcf, 0xcf, 0x3f, 0xff,
	0x00, 0xfc, 0xc3, 0xc3, 0x0f, 0x0f, 0xff, 0x00, 0xfc, 0x07, 0xe0, 0x1f, 0x80, 0xff, 0x00,
0xfc,
	0x1f, 0xf8, 0x7f, 0xe0, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc, 0xff, 0xff,
```

```
	0xff, 0xff, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff,
	0xff, 0x00, 0xf8, 0xff, 0xff, 0xff, 0xff, 0x7f, 0x00, 0xf8, 0xff, 0xff, 0xff, 0xff, 0x7f, 0x00,
	0xf8, 0xff, 0x07, 0x80, 0xff, 0x7f, 0x00, 0xf0, 0xff, 0x00, 0x00, 0xfc, 0x3f, 0x00, 0xf0,
0xff,
	0xf0, 0x3f, 0xfc, 0x3f, 0x00, 0xe0, 0xff, 0xff, 0xff, 0xff, 0x1f, 0x00, 0xe0, 0xff, 0xff, 0xff,
	0xff, 0x1f, 0x00, 0xc0, 0xff, 0xff, 0xff, 0xff, 0x0f, 0x00, 0xc0, 0xff, 0xff, 0xff, 0xff, 0x0f,
	0x00, 0x80, 0xff, 0xff, 0xff, 0xff, 0x07, 0x00, 0x00, 0xff, 0xff, 0xff, 0xff, 0x03, 0x00,
0x00,
	0xfe, 0xff, 0xff, 0xff, 0x01, 0x00, 0x00, 0xfc, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0xf8,
0xff,
	0xff, 0x7f, 0x00, 0x00, 0x00, 0xe0, 0xff, 0xff, 0x1f, 0x00, 0x00, 0x00, 0x80, 0xff, 0xff,
0x07,
	0x00, 0x00, 0x00, 0x00, 0xfe, 0xff, 0x01, 0x00, 0x00, 0x00, 0x00, 0xf0, 0x3f, 0x00,
0x00, 0x00,
	0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00
};

// 'hot', 50x50px
const unsigned char epd_bitmap_hot [] PROGMEM = {
	0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
0x00, 0x00,
	0xf0, 0x3f, 0x00, 0x00, 0x00, 0x00, 0x00, 0xfe, 0xff, 0x01, 0x00, 0x00, 0x00, 0x80, 0xff,
0xff,
	0x07, 0x00, 0x00, 0x00, 0xe0, 0xff, 0xff, 0x1f, 0x00, 0x00, 0x00, 0xf8, 0xff, 0xff, 0x7f,
0x00,
	0x00, 0x00, 0xfc, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0xfe, 0xff, 0xff, 0xff, 0x01, 0x00,
0x00,
	0xff, 0xff, 0xff, 0xff, 0x03, 0x00, 0x80, 0xff, 0xff, 0xff, 0xff, 0x07, 0x00, 0xc0, 0xff, 0xff,
	0xff, 0xff, 0x0f, 0x00, 0xc0, 0xff, 0xff, 0xff, 0xff, 0x0f, 0x00, 0xe0, 0x83, 0xff, 0xff,
0x07,
```

0x1f, 0x00, 0xe0, 0x03, 0xfe, 0xff, 0x01, 0x1f, 0x00, 0xf0, 0x3f, 0xf8, 0x7f, 0xf0, 0x3f, 0x00,

0xf0, 0xff, 0xf1, 0x3f, 0xfe, 0x3f, 0x00, 0xf8, 0xff, 0xe3, 0x1f, 0xff, 0x7f, 0x00, 0xf8, 0xff,

0xcf, 0xcf, 0xff, 0x7f, 0x00, 0xf8, 0xff, 0x8f, 0xc7, 0xff, 0x7f, 0x00, 0xfc, 0x3f, 0x9c, 0xe7,

0xf0, 0xff, 0x00, 0xfc, 0x3f, 0xf8, 0x7f, 0xf0, 0xff, 0x00, 0xfc, 0x3f, 0xf0, 0x3f, 0xf0, 0xff,

0x00, 0xfc, 0x3f, 0xf0, 0x3f, 0xf0, 0xff, 0x00, 0xfc, 0x7f, 0xf0, 0x3f, 0xf8, 0xff, 0x00, 0xfc,

0xff, 0xf0, 0x3f, 0xfc, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc, 0xff, 0xff,

0xff, 0xff, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff, 0xff, 0x00, 0xfc, 0xff, 0xff, 0xff, 0xff,

0xff, 0x00, 0xf8, 0xff, 0xff, 0xff, 0xff, 0x7f, 0x00, 0xf8, 0x3f, 0x0c, 0xc3, 0xf0, 0x7f, 0x00,

0xf8, 0x3f, 0x0c, 0xc3, 0xf0, 0x7f, 0x00, 0xf0, 0x7f, 0x0c, 0xc3, 0xf8, 0x3f, 0x00, 0xf0, 0x7f,

0x0c, 0xc3, 0xf8, 0x3f, 0x00, 0xe0, 0xff, 0x0c, 0xc3, 0xfc, 0x1f, 0x00, 0xe0, 0xff, 0x0d, 0xc3,

0xfe, 0x1f, 0x00, 0xc0, 0xff, 0x0f, 0xc3, 0xff, 0x0f, 0x00, 0xc0, 0xff, 0x1f, 0xe3, 0xff, 0x0f,

0x00, 0x80, 0xff, 0xff, 0xff, 0xff, 0x07, 0x00, 0x00, 0xff, 0xff, 0xff, 0xff, 0x03, 0x00, 0x00,

0xfe, 0xff, 0xff, 0xff, 0x01, 0x00, 0x00, 0xfc, 0xff, 0xff, 0xff, 0x00, 0x00, 0x00, 0xf8, 0xff,

0xff, 0x7f, 0x00, 0x00, 0x00, 0xe0, 0xff, 0xff, 0x1f, 0x00, 0x00, 0x00, 0x80, 0xff, 0xff, 0x07,

0x00, 0x00, 0x00, 0x00, 0xfe, 0xff, 0x01, 0x00, 0x00, 0x00, 0x00, 0xf0, 0x3f, 0x00, 0x00, 0x00,

0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00

};

```
DHT dht(DHT_SENSOR_PIN, DHTTYPE);

void setup() {
  // put your setup code here, to run once:
  Serial.begin(115200);  // Initialize serial communication
  Wire.begin(SDA_OLED, SCL_OLED);  // Initialize I2C communication
  OLED.begin();
  OLED.setFont(u8g2_font_t0_12b_mf);
  OLED.setColorIndex(1);
   pinMode(SOIL_MOISTURE_SENSOR_PIN, INPUT);   // Set the soil moisture sensor pin as
input
  pinMode(LDR_SENSOR_PIN, INPUT);  // Set the LDR sensor pin as input
  dht.begin();  // Initialize the DHT sensor
}

void loop() {
  int ldrValue = digitalRead(LDR_SENSOR_PIN);
  int sensorValue = digitalRead(SOIL_MOISTURE_SENSOR_PIN);
  float temperature = dht.readTemperature();

  // Print sensor status to the serial monitor
  Serial.print("LDR Value: ");
  Serial.println(ldrValue);
  Serial.print("Soil Moisture Value: ");
  Serial.println(sensorValue);
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");
```

```
if (isnan(temperature)) {
  Serial.println("Failed to read from DHT sensor!");
  return;
}

if (temperature > 30) {
  // Hot Face
  OLED.firstPage();
  do {
    OLED.drawXBMP(39, 7, 50, 50, epd_bitmap_hot);
  } while (OLED.nextPage());
} else if (ldrValue == LOW) {
  // Night time, display sleep image
  OLED.firstPage();
  do {
    OLED.drawXBMP(39, 7, 50, 50, epd_bitmap_sleep);
  } while (OLED.nextPage());
} else {
  // Day time, check soil moisture
  if (sensorValue == LOW) {
    // Happy Face
    OLED.firstPage();
    do {
      OLED.drawXBMP(39, 7, 50, 50, epd_bitmap_happy);
    } while (OLED.nextPage());
  } else {
    // Sad Face
    OLED.firstPage();
    do {
      OLED.drawXBMP(39, 7, 50, 50, epd_bitmap_sad);
```

```
    } while (OLED.nextPage());
   }
  }

  delay(1000);  // Update the display every second
}
```